Supplementary Materials For *Beyond Unitary Designs: Quantifying Trainability in Parameterized Quantum Circuits*

I. Preliminaries on Symbolic Integrals and Zonal Orthogonal Polynomials

A. Weingarten Calculus Theorem

In this supplementary material, we present the formalism for integrating monomials in the matrix elements of unitary matrices over the unitary group. This technique, rooted in random matrix theory and representation theory, was first explored by Weingarten in the context of asymptotic expansions [15], and subsequently refined by Collins, who obtained an exact combinatorial expression [3, 4]. As our subsequent derivations rely heavily on this result, we formulate it precisely below.

Theorem S.1. *Let $U(d)$ be the unitary group of dimension d, $I = i_1, \cdots, i_n$, $J = j_1, \cdots, j_n$ be two n-tuples of positive integers chosen from $\{1, 2, \cdots, n\}$, $I' = i'_1, \cdots, i'_m$, $J' = j'_1, \cdots, j'_m$ be two m-tuples of positive integers chosen from $\{1, 2, \cdots, n\}$. Then if $m = n$,*

$$\int_{U(d)} \mathrm{d}\mu_H(U) U_{IJ} U^*_{I'J'} = \sum_{\sigma,\tau\in S_n} \mathrm{Wg}(\tau^{-1}\sigma, d)\delta_{I,I'_\sigma}\delta_{J,J'_\tau}, \tag{S1}$$

here $I'_\sigma = i'_{\sigma(1)} i'_{\sigma(2)} \cdots i'_{\sigma(m)}$ and $J'_\tau = j'_{\tau(1)} j'_{\tau(2)} \cdots j'_{\tau(m)}$. If $m \neq n$, then

$$\int_{U(d)} \mathrm{d}\mu_H(U) U_{IJ} U^*_{I'J'} = 0. \tag{S2}$$

Here, the Weingarten function is defined by

$$\mathrm{Wg}(\tau^{-1}\sigma, d) = \frac{1}{n!}\sum_\lambda \frac{d_\lambda}{s_{\lambda,d}(1)}\chi_\lambda(\tau^{-1}\sigma), \tag{S3}$$

in which the sum is over all partitions λ of n, d_λ is the dimension of an irreducible representation represented by λ, and $s_{\lambda,d}(1)$ is the Schur polynomial evaluated at the identity.

B. Majority Identity for Variance Calculation

In this subsection, we list several key identities that recur throughout our computations, all of which follow from the Weingarten integration formalism outlined in Theorem S.1. These formulas facilitate the evaluation of moments and traces in unitary ensembles, with applications in the unitary design theory.

1. trace integral of 1 degree:

$$\int_{U(d)} \mathrm{d}\mu_H(U)\,\mathrm{Tr}[UAU^\dagger B] = \frac{\mathrm{Tr}[A]\,\mathrm{Tr}[B]}{d}. \tag{S4}$$

2. trace integral of 2 degree:

$$\begin{aligned}&\int_{U(d)} \mathrm{d}\mu_H(U)\,\mathrm{Tr}[UAU^\dagger BUCU^\dagger D]\\ &\qquad= \frac{\mathrm{Tr}[A]\,\mathrm{Tr}[C]\,\mathrm{Tr}[BD] + \mathrm{Tr}[AC]\,\mathrm{Tr}[B]\,\mathrm{Tr}[D]}{d^2-1} - \frac{\mathrm{Tr}[AC]\,\mathrm{Tr}[BD] + \mathrm{Tr}[A]\,\mathrm{Tr}[B]\,\mathrm{Tr}[C]\,\mathrm{Tr}[D]}{d(d^2-1)},\end{aligned} \tag{S5}$$

and

$$\begin{aligned}&\int_{U(d)} \mathrm{d}\mu_H(U)\,\mathrm{Tr}[UAU^\dagger B]\,\mathrm{Tr}[UCU^\dagger D]\\ &\qquad= \frac{\mathrm{Tr}[AC]\,\mathrm{Tr}[BD] + \mathrm{Tr}[A]\,\mathrm{Tr}[B]\,\mathrm{Tr}[C]\,\mathrm{Tr}[D]}{d^2-1} - \frac{\mathrm{Tr}[A]\,\mathrm{Tr}[C]\,\mathrm{Tr}[BD] + \mathrm{Tr}[AC]\,\mathrm{Tr}[B]\,\mathrm{Tr}[D]}{d(d^2-1)}.\end{aligned} \tag{S6}$$

3. Summation formula:

$$\sum_{\sigma\in S_n} \mathrm{Wg}(\sigma, d) = \frac{(d-1)!}{(n+d-1)!}. \tag{S7}$$

4. n-th moment of a single unitary matrix element with a bias:

$$\int d\mu_H(U)|U_{ij}|^{2n}|U_{ik}|^2 = \begin{cases} \dfrac{n!(d-1)!}{(n+d)!}, & \text{if } j \neq k \\ \dfrac{(n+1)!(d-1)!}{(n+d)!}, & \text{if } j = k \end{cases}. \tag{S8}$$

Equations (S4) and (S6) are standard results in the literature (see, e.g., [8, 11]), while (S7) is a well-established identity in representation theory [3]. For completeness, we derive (S8) via Theorem S.1.

Proof of Eq.(S8). First, consider the case $j = k$:

$$\int d\mu_H(U)|U_{ij}|^{2(n+1)} = \sum_{\sigma,\tau\in S_{n+1}} \mathrm{Wg}(\sigma\tau^{-1}, d) = (n+1)! \cdot \sum_{\sigma} \mathrm{Wg}(\sigma, d) = \frac{(n+1)!(d-1)!}{(n+d)!}. \tag{S9}$$

The second equality arises from the invariance properties of the Weingarten function under group averaging, and the final step invokes (S7) (noting the shift $n \to n+1$). Next, for $j \neq k$, Theorem S.1 and (S7) yields:

$$\int d\mu_H(U)|U_{ij}|^{2n}|U_{ik}|^2 = \sum_{\sigma\in S_{n+1},\tau\in S_n} \mathrm{Wg}(\sigma(\tau 1)^{-1}, d) = n! \sum_{\sigma\in S_{n+1}} \mathrm{Wg}(\sigma, d) = \frac{n!(d-1)!}{(n+d)!}. \tag{S10}$$

Here, $(\tau\, 1)$ denotes the natural embedding of $\tau \in S_n$ into S_{n+1} by fixing the point $n+1$. □

C. Evaluation of the nth Moment of $\mathrm{Tr}[U\rho U^\dagger X]^2$

In this subsection, we compute the moments of the random variable $\mathrm{Tr}[U\rho U^\dagger X]^2$ under the Haar measure on $U(d)$. We assume that ρ is a pure state and that X is a traceless Hermitian matrix satisfying $X^2 = I_d$ (e.g., a Pauli word).

Proposition S.1. *For pure ρ and $n \in \mathbb{N}_+$, the n-th moment of $\mathrm{Tr}[U\rho U^\dagger X]^2$ is*

$$\| \mathrm{Tr}[U\rho U^\dagger X]^2 \|_n^n = \frac{(d-1)!}{(2n+d-1)!} \sum_{\sigma\in S_{2n}} \mathrm{Tr}[X^{\lambda_1^\sigma}]\, \mathrm{Tr}[X^{\lambda_2^\sigma}] \cdots \mathrm{Tr}[X^{\lambda_m^\sigma}], \tag{S11}$$

where the sum runs over all permutations $\sigma \in S_{2n}$, and $(\lambda_1^\sigma, \lambda_2^\sigma, \cdots, \lambda_m^\sigma)$ is the partition of σ.

Proof. We begin by expressing the integral in index notation:

$$\begin{aligned} \int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger X]^{2n} &= \int \mathrm{d}\mu(U) \prod_{s=1}^{2n} \sum_{i_s, j_s, i'_s, j'_s} \left(U_{i_s j_s} \rho_{j_s j'_s} U^*_{i'_s j'_s} X_{i'_s i_s} \right) \\ &= \int \mathrm{d}\mu_H(U) \prod_{s=1}^{2n} \sum_{i_s, j_s, i'_s, j'_s} \left(U_{i_s j_s} \rho_{j_s j'_s} U^*_{i'_s j'_s} X_{i'_s i_s} \right). \end{aligned} \tag{S12}$$

Due to the left-invariance of the Haar measure, i.e., $\mu_H(S) = \mu_H(gS)$ for any $g \in U(d)$ and measurable subset $S \subset U(d)$, we

may assume without loss of generality that $\rho = \mathrm{diag}(1, 0, \ldots, 0)$. Applying Theorem S.1 and Equation (S7), we obtain

$$\begin{aligned}
\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger X]^{2n} &= \int \mathrm{d}\mu_H(U) \prod_{s=1}^{2n} \sum_{i_s, i'_s} \left(U_{i_s 1} U^*_{i'_s 1} X_{i'_s i_s} \right) \\
&= \sum \int \mathrm{d}\mu_H(U) U_{i_1 1} U^*_{i'_1 1} U_{i_2 1} U^*_{i'_2 1} \cdots U_{i_{2n} 1} U^*_{i'_{2n} 1} X_{i'_1 i_1} X_{i'_2 i_2} \cdots X_{i'_{2n} i_{2n}} \\
&= \sum \sum_{\sigma, \tau \in S_{2n}} \mathrm{Wg}(\tau^{-1}\sigma, d) \delta_{i_1, i'_{\sigma(1)}} \cdots \delta_{i_{2n}, i'_{\sigma(2n)}} X_{i'_1 i_1} X_{i'_2 i_2} \cdots X_{i'_{2n} i_{2n}} \\
&= \sum \sum_{\sigma \in S_{2n}} \frac{(d-1)!}{(2n+d-1)!} \delta_{i_1, i'_{\sigma(1)}} \cdots \delta_{i_{2n}, i'_{\sigma(2n)}} X_{i'_1 i_1} X_{i'_2 i_2} \cdots X_{i'_{2n} i_{2n}} \\
&= \frac{(d-1)!}{(2n+d-1)!} \sum_{\sigma \in S_{2n}} \mathrm{Tr}[X^{\lambda_1^\sigma}]\,\mathrm{Tr}[X^{\lambda_2^\sigma}] \cdots \mathrm{Tr}[X^{\lambda_m^\sigma}].
\end{aligned} \tag{S13}$$

This completes the proof. □

We now impose the additional assumption that $|\,\mathrm{Tr}[X^k]| \le C$ for all $k \ge 1$ and some constant $C > 0$. Then,

$$\|\,\mathrm{Tr}[U\rho U^\dagger X]^2\|_n^n \lesssim \frac{(d-1)!}{(2n+d-1)!} |S|. \tag{S14}$$

where $S \subseteq S_{2n}$ is the subset of permutations without 1-cycles. The cardinality $|S|$ can be computed using combinatorial methods, yielding that $|S| = (2n)! \sum_{k=0}^{2n} \frac{(-1)^k}{k!}$. Thus,

$$\|\,\mathrm{Tr}[U\rho U^\dagger X]^2\|_n \lesssim \left(\frac{(d-1)!(2n)!}{(2n+d-1)!} \sum_{k=0}^{2n} \frac{(-1)^k}{k!} \right)^{1/n}. \tag{S15}$$

Noting that $\sum_{k=0}^{2n} \frac{(-1)^k}{k!} \le 1$, we have

$$\|\,\mathrm{Tr}[U\rho U^\dagger X]^2\|_n \lesssim \binom{2n+d-1}{d-1}^{-1/n}. \tag{S16}$$

II. Preliminaries on Zonal Orthogonal Polynomials

In this section, we present the foundational concepts of zonal orthogonal polynomials. Specifically, we discuss the properties of partial orthogonality and completeness, which form the mathematical foundation of our framework.

Any irreducible representation of the unitary group $U(d)$ can be labeled by a non-increasing integer sequence $\boldsymbol{\lambda} := (\lambda_1, \lambda_2, \cdots, \lambda_d)$ of length d, where $\lambda_i \in \mathbb{Z}$ and $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_d$ [2]. For each irreducible representation $\boldsymbol{\lambda}$, one can define a family of polynomials $\{Z_{\lambda,V} : V \in U(d)\}$, known as zonal polynomials [5, 9, 10], which form a basis for the vector space of the balanced polynomials on $U(d)$.

A key property of zonal polynomials is their orthogonality across distinct indices [12],

$$\left\langle Z_{\lambda,V_1}, Z_{\lambda',V_2} \right\rangle = \delta_{\lambda,\lambda'} \left\langle Z_{\lambda,V_1}, Z_{\lambda,V_2} \right\rangle, \tag{S17}$$

where $\langle \cdot, \cdot \rangle$ is the Hilbert-Schmidt inner product, i.e. $\langle f, g \rangle = \int \mathrm{d}\mu_H \bar{f} g$. It is known that every element in $\mathrm{Hom}(U(d), t, t)$ can be expressed as a linear combination of zonal polynomials in $\{Z_{\lambda,U} : U \in U(d), |\lambda| = 0, |\boldsymbol{\lambda}_+| \le t\}$, where $\boldsymbol{\lambda}_+$ denotes the sum of positive entries of $\boldsymbol{\lambda}$ and $|\lambda|$ denotes the sum of all entries of $\boldsymbol{\lambda}$. Formally, any balanced polynomial of degree t admits the expansion

$$p = \sum_{|\lambda|=0, |\boldsymbol{\lambda}_+| \le t} \sum_{V_\lambda} a_{\lambda,V_\lambda} Z_{\lambda,V_\lambda}, \tag{S18}$$

where $a_{\lambda,V_\lambda} \in \mathbb{R}$ is the coefficient of each zonal polynomial Z_{λ,V_λ}. Since the variance function $\mathrm{Tr}[U\rho U^\dagger X]^2$ is a balanced polynomial with degree ≤ 2. For any polynomial p, if we focus on the integral

$$\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger X]^2 p(U) = \left\langle \mathrm{Tr}[\cdot \rho \cdot^\dagger X]^2, p \right\rangle, \tag{S19}$$

by the orthogonality, it suffices to restrict attention to those zonal polynomials with indices in $S_{\le 2} := \{\boldsymbol{\lambda} : |\lambda| = 0, |\boldsymbol{\lambda}_+| \le 2\}$. There are $|S_{\le 2}| = 6$ such types of zonal polynomials, whose explicit forms are detailed in Table S1.

TABLE S1: The list of zonal polynomials with degree at most 2

λ	$Z_{\lambda,V}(U)$
$(0,0,\cdots,0,0)$	1
$(1,0,\cdots,0,-1)$	$(d^2-1)(\lvert\operatorname{Tr}[V^\dagger U]\rvert^2-1)$
$(2,0,\cdots,0,-2)$	$\frac{d^2(d-1)(d+3)}{4}(\frac{1}{4}\lvert\operatorname{Tr}[V^\dagger U]^2+\operatorname{Tr}[V^\dagger UV^\dagger U]\rvert^2-\lvert\operatorname{Tr}[V^\dagger U]\rvert^2)$
$(2,0,\cdots,0,-1,-1)$	$\frac{(d^2-1)(d^2-4)}{4}\left(\frac{1}{4}(\operatorname{Tr}[V^\dagger U]^2+\operatorname{Tr}[V^\dagger UV^\dagger U])(\overline{\operatorname{Tr}[V^\dagger U]^2-\operatorname{Tr}[V^\dagger UV^\dagger U]})-\lvert\operatorname{Tr}[V^\dagger U]\rvert^2+1\right)$
$(1,1,0,\cdots,0,-2)$	$\frac{(d^2-1)(d^2-4)}{4}\left(\frac{1}{4}(\operatorname{Tr}[V^\dagger U]^2-\operatorname{Tr}[V^\dagger UV^\dagger U])(\overline{\operatorname{Tr}[V^\dagger U]^2+\operatorname{Tr}[V^\dagger UV^\dagger U]})-\lvert\operatorname{Tr}[V^\dagger U]\rvert^2+1\right)$
$(1,1,0,\cdots,0,-1,-1)$	$\frac{d^2(d+1)(d-3)}{4}(\frac{1}{4}\lvert\operatorname{Tr}[V^\dagger U]^2-\operatorname{Tr}[V^\dagger UV^\dagger U]\rvert^2-\lvert\operatorname{Tr}[V^\dagger U]\rvert^2)$

III. Approximation by Polynomial Measures

This section introduces the framework of standard distribution theory from [7], which provides a rigorous proof of our basic theoretical framework. Based on it, we prove that the distribution of any PQC can be approximated with arbitrary precision while preserving its gradient variance, which makes our framework applicable to the analysis of trainability. Before we formally discuss, some preliminaries about probability distribution and related approximation theorem need to be introduced.

Here, let G be a connected compact Lie group. Fix a real Borel measure μ on a connected subset $S \subseteq G$. Then we can define a distribution by

$$\begin{aligned} T_\mu : \mathcal{D}(G) &\longrightarrow \mathbb{R} \\ \phi &\longmapsto \int_S \phi|_S \,\mathrm{d}\mu \end{aligned}. \tag{S20}$$

Through this approach, we embed the space of Borel measures $\mathcal{M}(G)$ into the distribution space $\mathcal{D}'(G)$. We now introduce a fundamental concept in harmonic analysis, the approximation to the identity, which provides a standard framework for establishing function approximation via convolution.

Definition S.1 (Approximation to the identity)**.** *A family $\{\eta_\varepsilon\}_{\varepsilon>0} \subseteq C^\infty(G)$ is called an approximation to the identity if it satisfies the following conditions:*

1. *Non-negativity:*

$$\eta_\varepsilon \geq 0. \tag{S21}$$

2. *Normalization:*

$$\int \eta_\varepsilon \,\mathrm{d}\mu_H = 1, \quad \forall \varepsilon > 0. \tag{S22}$$

3. *Concentration at the identity:*

$$\lim_{U \ni e} \int_{G\setminus U} \eta_\varepsilon \,\mathrm{d}\mu_H = 0, \quad \forall \varepsilon > 0, \tag{S23}$$

where e is the identity of G and U is an open neighbourhood of e in G.

Indeed, one can always find a positive smooth function $\gamma \in C^\infty(G)$, such that the family $\{\gamma_\varepsilon := \varepsilon^{-1}\gamma(x/\varepsilon) | \varepsilon > 0\}$ is an approximation to the identity. Employing the definition of approximation to the identity, we have the following approximate theorem:

Theorem S.2. *Let $\{\eta_\varepsilon\}_{\varepsilon>0}$ be an approximation to the identity and $T \in \mathcal{D}'(G)$, then*

$$T * \eta_\varepsilon \to T \quad \text{as} \quad \varepsilon \to 0 \tag{S24}$$

*with respect to the weak *-topology. Here, the convolution $T * \eta_\varepsilon$ is defined by*

$$\begin{aligned} T * \eta_\varepsilon : \mathcal{D}(G) &\longrightarrow \mathbb{R} \\ \phi &\longmapsto \left\langle T, \eta_\varepsilon^\vee * \phi \right\rangle, \end{aligned} \tag{S25}$$

where $\eta_\varepsilon^\vee(g) = \eta_\varepsilon(g^{-1})$.

The distribution $T * \eta_\varepsilon$ is represented by a smooth function, implying that $C^\infty(G)$ is weak *-dense in $\mathcal{D}'(G)$. Since $C^\infty(G) \subseteq \mathcal{M}(G) \subseteq \mathcal{D}'(G)$, it follows that $C^\infty(G)$ is weak *-dense in $\mathcal{M}(G)$. For parametrized quantum circuits, the induced measures are probability measures, so we henceforth restrict to this case. Here, we introduce the definition of the probability distribution.

Definition S.2 (Probability distribution)**.** *A distribution $T \in \mathcal{D}'(G)$ is a probability distribution if it satisfies the following two conditions:*

1. *T is positive: $\langle T, \phi\rangle \geq 0$ whenever $\phi \geq 0$.*
2. *The value of T at 1 is 1: $T(1) = 1$.*

Here, we denote $\mathcal{D}'(G)^1$ as the set consisting of all the probability distributions on G.

Indeed, when considering a distribution induced by a Borel measure μ on the subset S, the statement that μ is a probability measure is equivalent to the statement that T_μ is a probability distribution. Here we denote $C^\infty(G)^1$ and $\mathcal{M}(G)^1$ to be the set of smooth functions with total mass 1 and the set of probability measures. Noted that if μ is a probability measure, then $T_\mu * \eta_\varepsilon \in C^\infty(G)^1$, we can find that $C^\infty(G)^1$ is weak *-dense in $\mathcal{M}(G)^1$.

Now return to our setting that $G = U(d)$. Through Theorem S.2, one can find a sequence $\{\zeta_n\}_{n=1}^\infty \subseteq C_c^\infty(U(d))$, such that $\zeta_n \overset{*}{\rightharpoonup} T$, equivalently, for any $\phi \in C_c^\infty(U(d))$, we have

$$\langle \zeta_n, \phi\rangle \to \langle T, \phi\rangle \,. \tag{S26}$$

We now introduce the concept of a weak *-open neighborhood as follows:

Definition S.3. *Let I be a finite indexing set and let $f_i \in \mathcal{D}(U(d)$ for $i \in I$. Let $\varepsilon > 0$. Then, a weak* open neighbourhood of T_0 can be written as*

$$\{T \in \mathcal{D}'(U(d)) : |\langle T, f_i\rangle - \langle T_0, f_i\rangle| < \varepsilon,\ \forall\, i \in I\}. \tag{S27}$$

Furthermore, for $T \in \mathcal{D}'(U(d))$, we define

$$\begin{aligned} f_T : U(d) &\longrightarrow \mathbb{R} \\ U &\longmapsto \left\langle T, \mathrm{Tr}[\cdot\rho \cdot^\dagger i[H_\ell, U^\dagger O U]\right\rangle . \end{aligned} \tag{S28}$$

We then have the following proposition:

Proposition S.2. *Consider the map on $\mathcal{D}'(U(d)) \times \mathcal{D}'(U(d))$ defined as follows:*

$$\begin{aligned} \mathrm{Var} : \mathcal{D}'(U(d)) \times \mathcal{D}'(U(d)) &\longrightarrow \mathbb{R} \\ (L, R) &\longmapsto L(f_R). \end{aligned} \tag{S29}$$

This map Var *is continuous.*

Proof. Given any pair of distributions (L, R) and $\varepsilon > 0$, it suffices to show that there exists an open neighborhood of (L, R) such that for any pair (L', R') in this neighborhood,

$$|\mathrm{Var}(L', R') - \mathrm{Var}(L, R)| < \varepsilon. \tag{S30}$$

Take an open neighborhood of L in $\mathcal{D}'(U(d))$ given by

$$S_L = \{L' \in \mathcal{D}'(U(d)) : |\langle L, f_R\rangle - \langle L', f_R\rangle| < \frac{1}{2}\varepsilon\} \tag{S31}$$

and an open neighbourhood of R in $\mathcal{D}'(U(d))$ by

$$S_R = \bigcap_{\phi \in \mathcal{D}(U(d))} \{L' \in \mathcal{D}'(U(d)) : |\langle R, \phi\rangle - \langle R', \phi\rangle| < \frac{1}{2}\varepsilon\}. \tag{S32}$$

Then, for any pair (L', R) lying in the neighbourhood $S_L \times S_R$, we have

$$|\mathrm{Var}(L', R') - \mathrm{Var}(L, R)| \leq |L'(f_{R'}) - L'(f_R)| + |L'(f_R) - L(f_R)|. \tag{S33}$$

Since $L' \in S_L$, we have $|L'(f_R) - L(f_R)| < 1/2\varepsilon$. And $R' \in S_R$, we have

$$|f_{R'} - f_R| = |\left\langle R', \mathrm{Tr}[\cdot\rho \cdot^\dagger i[H_\ell, U^\dagger O U]\right\rangle - \left\langle R, \mathrm{Tr}[\cdot\rho \cdot^\dagger i[H_\ell, U^\dagger O U]\right\rangle| < 1/2\varepsilon. \tag{S34}$$

Thus, $|\mathrm{Var}(L', R') - \mathrm{Var}(L, R)| < \varepsilon$. □

Indeed, the composition map of the embedding map from $\mathcal{M}(U(d))$ to $\mathcal{D}'(U(d))$ and Var is the same as the usual variance calculation formula under the zero-bias network assumption:

$$\begin{aligned}\mathrm{Var}' : \mathcal{M}(U(d)) \times \mathcal{M}(U(d)) &\longrightarrow \mathbb{R} \\ (\mu_L, \mu_R) &\longmapsto \int \mathrm{d}\mu_L(U_L)\, \mathrm{d}\mu_R(U_R)\, \mathrm{Tr}[U_R \rho U_R^\dagger \left(i[H_\ell, U_L^\dagger O U_L]\right)]^2.\end{aligned} \tag{S35}$$

Clearly, this map Var′ is also continuous, which leads to the compatibility between variance approximation and the approximation of measures with respect to the weak *-topology.

A. A Trick to restrict to Balanced Polynomial Measures

In the main text, the approximation framework reduces the evaluation of gradient variances for arbitrary PQCs to polynomial integrals over the Haar measure. By Theorem S.1, nonbalanced terms vanish, allowing one to restrict to balanced polynomial measures. Moreover, there exists an alternative and more direct approach that approximates arbitrary PQCs using balanced polynomial measures.

Here, we introduce two global phase factors $e^{i\alpha}$ and $e^{i\beta}$, where α and β are independent and uniformly distributed on $[0, 2\pi)$, and we define $U_L' := e^{i\alpha} U_L$ and $U_R' := e^{i\beta} U_R$, respectively. It leads to the conclusion that the corresponding probability measures μ_L and μ_R become invariant under the scalar action of the unit circle $\mathbb{S}^1$. Moreover, the variance is unchanged under the transformation, i.e. $\mathrm{Var}_\theta^{(\mu_L', \mu_R')}[\partial_\ell C] = \mathrm{Var}_\theta^{(\mu_L, \mu_R)}[\partial_\ell C]$, since the partial derivative $\partial_\ell C = i\,\mathrm{Tr}[U_R \rho U_R^\dagger [H_\ell, U_L^\dagger O U_L]]$ is invariant under transformations $U_R \to e^{i\alpha} U_R$ and $U_L \to e^{i\beta} U_L$. Therefore, we can make the PQC $\mathbb{S}^1$-invariant by introducing global factors without affecting the variance of the gradient at any parameter. Furthermore, we have the following proposition:

Proposition S.3. *Any complex-valued polynomial on $U(d)$ that is invariant under the $\mathbb{S}^1$-action must be a balanced polynomial. Formally,*

$$\mathcal{I}(U(d))^{\mathbb{S}^1} = \mathcal{I}_B(U(d)). \tag{S36}$$

As a corollary, $\mathcal{I}_B(U(d))$ is dense in $\mathcal{P}(U(d)))^{\mathbb{S}^1}$.

Proof. Given a polynomial p on $U(d)$, one can decompose it into the sum of homogeneous polynomials,

$$p(U) = \sum_{t,r} p_{(t,r)}(U), \tag{S37}$$

where $p_{(t,r)}$ is the homogeneous polynomial of degree (t, s). Indeed, p is $\mathbb{S}^1$-invariant iff all of the $p_{(t,r)}$ are $\mathbb{S}^1$-invariant since $P_{(t,r)}$ are independent of each other. Noted that $p_{(t,r)}(sU) = s^{t-r} p_{(t,r)}(U)$ for $s \in \mathbb{S}^1$, nonconstant $p_{(t,r)}$ is $\mathbb{S}^1$-invariant iff $t = r$. Thus, we obtain

$$\mathcal{I}(U(d))^{\mathbb{S}^1} = \mathcal{I}_B(U(d)). \tag{S38}$$

Since $\mathcal{I}(U(d))$ is dense in $\mathcal{P}^q(U(d)))$ for any $1 \le q \le \infty$, $\mathcal{I}_B(U(d)) = \mathcal{I}(U(d))^{\mathbb{S}^1}$ is dense in $\mathcal{P}^q(U(d)))^{\mathbb{S}^1}$ with respect to the subspace topology. □

Proposition S.3 shows that any $\mathbb{S}^1$-invariant PQC can be approximated by balanced polynomial measures. Moreover, since any PQC can be made $\mathbb{S}^1$-invariant by introducing global phase factors, which do not affect the gradient variance, this yields a direct approximation of arbitrary PQCs by balanced polynomial measures.

B. Proof of Theorem 1

Theorem 1. *Let $R = \exp(iP\theta)$ be a rotation gate, where $P = \otimes_{i=1}^n \sigma_i$ is a Pauli word and $\theta \sim \mathrm{Uniform}[0, 2\pi)$. The distribution of R admits an approximation by the following partial sums*

$$p_N = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \sum_{|\lambda| \le N} Z_{\lambda, e^{iP\theta}}, \quad N \in \mathbb{Z}_+. \tag{S39}$$

Furthermore, at the level of computing gradient variance, the distribution of R is equivalent to

$$p_{2,b} = 1 + \frac{1}{2}(Z_{(1,0,\cdots,0,-1),I} + Z_{(1,0,\cdots,0,-1),P}) + \frac{1}{4}\sum_{|\lambda|=0,\ |\lambda_+|=2}(Z_{\lambda,I} + Z_{\lambda,P} + Z_{\lambda,\frac{(I+iP)}{\sqrt{2}}} + Z_{\lambda,\frac{(I-iP)}{\sqrt{2}}}). \tag{S40}$$

For a fixed entangling gate W_ℓ, its distribution is given by the Dirac delta distribution concentrated at W_ℓ.

Proof. By the Peter-Weyl theorem, for any $U \in U(d)$,

$$\delta_U = \sum_\lambda Z_{\lambda,U} \tag{S41}$$

with respect to the weak *-topology. Let E denote the set $\{e^{iP\theta} : \theta \in [0, 2\pi)\}$, then the distribution δ_E defined by

$$\begin{aligned} \delta_E : \mathcal{D}(G) &\longrightarrow \mathbb{R} \\ f &\longmapsto \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta f(e^{iP\theta}) \end{aligned} \tag{S42}$$

can be written as a power series

$$\delta_E = \sum_\lambda \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta Z_{\lambda,e^{iP\theta}}. \tag{S43}$$

Noted that the variance function $\mathrm{Tr}[U\rho U^\dagger X_L]^2$ is a balanced polynomial with respect to U with degree 2, it can be decomposed into

$$\mathrm{Tr}[U\rho U^\dagger X_L]^2 = \sum_{|\lambda|=0,\ |\lambda_+|\le 2}\sum_{U'_\lambda} a_{\lambda,U'_\lambda} Z_{\lambda,U'_\lambda}. \tag{S44}$$

Therefore, when calculating the gradient variance, only $\sum_{|\lambda|=0,\ |\lambda_+|\le 2}\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{\lambda,e^{iP\theta}}$ contributes. For the case $\boldsymbol{\lambda} = (0, \cdots, 0)$,

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{(0,\cdots,0),e^{iP\theta}} = \frac{1}{2\pi}\int_0^{2\pi} 1\,\mathrm{d}\theta = 1. \tag{S45}$$

For the case $\boldsymbol{\lambda} = (1, 0, \cdots, 0, -1)$,

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{(1,0,\cdots,0,-1),e^{iP\theta}}(U) = (d^2-1)\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta\left(|\mathrm{Tr}[e^{iP\theta}U]|^2 - 1\right). \tag{S46}$$

Since P is Pauli word, $\exp(iP\theta) = \cos(\theta)I + i\sin(\theta)P$, which leads to

$$\begin{aligned} \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta|\mathrm{Tr}[e^{iP\theta}U]|^2 &= \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta\left(\cos^2(\theta)|\mathrm{Tr}[U]|^2 + \sin^2(\theta)|\mathrm{Tr}[PU]|^2\right) \\ &= \frac{1}{2}\left(|\mathrm{Tr}[U]|^2 + |\mathrm{Tr}[PU]|^2\right). \end{aligned} \tag{S47}$$

Thus,

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{(1,0,\cdots,0,-1),e^{iP\theta}} = \frac{1}{2}\left(Z_{(1,0,\cdots,0,-1),I} + Z_{(1,0,\cdots,0,-1),P}\right). \tag{S48}$$

For the case $\boldsymbol{\lambda} = (2, 0, \cdots, 0, -2)$,

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{(2,0,\cdots,0,-2),e^{iP\theta}}(U) = \frac{d^2(d-1)(d+3)}{4}\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta\left(\frac{1}{4}|\mathrm{Tr}[e^{iP\theta}U]^2 + \mathrm{Tr}[e^{iP\theta}Ue^{iP\theta}U]|^2 - |\mathrm{Tr}[e^{iP\theta}U]|^2\right). \tag{S49}$$

Integrating by θ, we can obtain

$$\begin{aligned} \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta\left(|\mathrm{Tr}[e^{iP\theta}U]^2 + \mathrm{Tr}[e^{iP\theta}Ue^{iP\theta}U]|^2\right) = \frac{|\mathrm{Tr}^2(V) + \mathrm{Tr}(V^2)|^2 + |\mathrm{Tr}^2(PV) + \mathrm{Tr}((PV)^2)|^2}{4} + \\ \frac{|\mathrm{Tr}^2(U) + \mathrm{Tr}(U^2) - \mathrm{Tr}^2(PU) - \mathrm{Tr}((PU)^2)|^2}{8} + \frac{|\mathrm{Tr}(PU^2) + \mathrm{Tr}(U)\,\mathrm{Tr}(PU)|^2}{2}. \end{aligned} \tag{S50}$$

Observed that

$$\frac{|\operatorname{Tr}^2(PU)-\operatorname{Tr}((PU)^2)|^2}{8}+\frac{|\operatorname{Tr}(PU^2)+\operatorname{Tr}(U)\operatorname{Tr}(PU)|^2}{2}=\frac{|\operatorname{Tr}^2(\frac{I+iP}{\sqrt{2}}U)+\operatorname{Tr}((\frac{I+iP}{\sqrt{2}}U)^2)|^2+|\operatorname{Tr}^2(\frac{I-iP}{\sqrt{2}}U)+\operatorname{Tr}((\frac{I-iP}{\sqrt{2}}U)^2)|^2}{4} \tag{S51}$$

and

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta|\operatorname{Tr}[e^{iP\theta}U]|^2=\frac{1}{2}\left(|\operatorname{Tr}[\frac{I+iP}{\sqrt{2}}U]|^2+|\operatorname{Tr}[\frac{I-iP}{\sqrt{2}}U]|^2\right), \tag{S52}$$

we obtain the exact expression that

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{(2,0,\cdots,0,-2),e^{iP\theta}}=\frac{1}{4}\left(Z_{(2,0,\cdots,0,-2),I}+Z_{(2,0,\cdots,0,-2),P}+Z_{(2,0,\cdots,0,-2),(I+iP)/\sqrt{2}}+Z_{(2,0,\cdots,0,-2),(I-iP)/\sqrt{2}}\right), \tag{S53}$$

Similarly, for the case $\boldsymbol{\lambda}=(2,0,\cdots,0,-1,-1),\ (1,1,0,\cdots,0,-2),\ (1,1,0,\cdots,0,-1,-1)$, there exists the similarly expressions that

$$\frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\theta Z_{\lambda,e^{iP\theta}}=\frac{1}{4}\left(Z_{\lambda,I}+Z_{\lambda,P}+Z_{\lambda,(I+iP)/\sqrt{2}}+Z_{\lambda,(I-iP)/\sqrt{2}}\right). \tag{S54}$$

□

C. Measures in Sequential Connection and Proof of Corollary 1.1

For two blocks U_1 and U_2 governed by the probability measures μ_1 and μ_2, respectively, where the output of U_1 passes to the input of U_2, the composite block U_1U_2 in a sequential arrangement follows the convolution of the measures $\mu_1*\mu_2$. The convolution of measures is defined as follows:

Definition S.4 (Convolution of measures)**.** *Given two finite measures μ_1 and μ_2 on the unitary group $U(d)$, the convolution $\mu_1*\mu_2$ is defined by*

$$\mu_1*\mu_2(S):=\int_{V\in S}f_1(U)f_2(U^\dagger V)d\mu_H(U).$$

for any measurable subset $S\subseteq U(d)$, where $f_1=\frac{d\mu_1}{d\mu_H}$ and $f_2=\frac{d\mu_2}{d\mu_H}$ are the Radon-Nikodym derivatives of μ_1 and μ_2 with respect to the standard Haar measure μ_H , respectively. Moreover, if μ_1 and μ_2 are probability measures, then $\mu_1*\mu_2$ is also a probability measure.

Notably, $\mu_1*\mu_2$ does not identify with $\mu_2*\mu_1$, because $U(d)$ is not abelian. And if μ_2 is a balanced polynomial measure, $\mu_1*\mu_2$ is also a balanced polynomial measure with $\deg(\mu_1*\mu_2)\le\deg(\mu_2)$.

From Definition S.4, the probability measure of the whole circuit is just the convolution of the probability measures of each component in order. Thus, we can summarize the following corollary:

Corollary 1.1. *Let $U(\theta)=\prod_{j=1}^L\exp(iP_j\theta_j)W_j$ be a PQC, where P_j are Pauli words and W_j are fixed unitary operators. Then the distribution of U can be equivalent to the following polynomials when calculating the gradient variance*

$$1+\frac{1}{2^L}\sum_{i_1,\ldots,i_L=1}^{2}Z_{(1,0,\ldots,0,-1),\prod_{s=1}^L P_s^{i_s}W_s}+\frac{1}{4^L}\sum_{|\lambda|=0,\ |\lambda_+|=2}\sum_{i_1,\ldots,i_L=1}^{4}Z_{\lambda,\prod_{s=1}^L P_s^{i_s}W_s}. \tag{S55}$$

Here, for a Pauli word P, we write $P^1:=I,\ P^2:=P,\ P^3:=(I+iP)/\sqrt{2},\ P^4:=(I-iP)/\sqrt{2}$.

Proof. For the parameterized quantum circuit blocks U and V, whose corresponding probability distributions are f and g respectively, the probability distribution corresponding to the parameterized quantum circuit UV is $f*g$. Therefore, letting $U_j=\exp(iP_j\theta_j)$, it can be obtained that when θ_j follows a uniform distribution over $[0,2\pi)$, the probability distribution corresponding to U_jW_j is

$$\begin{aligned}\frac{1}{2\pi}\int\mathrm{d}\theta_j\left(\sum_\lambda Z_{\lambda,U_j}*\delta_{W_j}\right)(V)&=\frac{1}{2\pi}\int\mathrm{d}\theta_j\sum_\lambda\int\mathrm{d}\mu_H(U)Z_{\lambda,U_j}(U)\delta_{W_j}(UV^\dagger)\\&=\frac{1}{2\pi}\int\mathrm{d}\theta_j\sum_\lambda Z_{\lambda,U_j}(VW_j^\dagger)\\&=\frac{1}{2\pi}\int\mathrm{d}\theta_j\sum_\lambda Z_{\lambda,U_jW_j}(V).\end{aligned} \tag{S56}$$

On this basis, the probability distribution corresponding to $U_j W_j U_{j+1}$ can be further computed as

$$\begin{aligned}\left(\frac{1}{2\pi}\right)^2 \int \mathrm{d}\theta_j \mathrm{d}\theta_{j+1} \left(\sum_{\lambda} Z_{\lambda,U_j W_j} * \sum_{\nu} Z_{\nu,U_{j+1}}\right)(V) &= \left(\frac{1}{2\pi}\right)^2 \int \mathrm{d}\theta_j \mathrm{d}\theta_{j+1} \sum_{\lambda,\nu} \int \mathrm{d}\mu(U) Z_{\lambda,U_j W_j}(U) Z_{\nu,U_{j+1}}(U^\dagger V) \\ &= \left(\frac{1}{2\pi}\right)^2 \int \mathrm{d}\theta_j \mathrm{d}\theta_{j+1} \sum_{\lambda,\nu} \left\langle Z_{\nu,VU_{j+1}^\dagger}, Z_{\lambda,U_j W_j} \right\rangle \\ &= \left(\frac{1}{2\pi}\right)^2 \int \mathrm{d}\theta_j \mathrm{d}\theta_{j+1} \sum_{\lambda} Z_{\lambda,U_j W_j}(VU_{j+1}^\dagger) \\ &= \left(\frac{1}{2\pi}\right)^2 \int \mathrm{d}\theta_j \mathrm{d}\theta_{j+1} \sum_{\lambda} Z_{\lambda,U_j W_j U_{j+1}}(V).\end{aligned} \tag{S57}$$

By analogy, the probability distribution corresponding to $U(\theta)$ can be obtained as

$$\left(\frac{1}{2\pi}\right)^L \int \prod_{j=1}^{L} \mathrm{d}\theta_j \sum_{\lambda} Z_{\lambda,U}. \tag{S58}$$

When computing the variance of the gradient, only the contributions from the polynomials in the region where the indices belong to $\boldsymbol{\lambda} : |\boldsymbol{\lambda}| = 0, |\boldsymbol{\lambda}_+| \leq 2$ are relevant. Therefore, based on the conclusion given in Theorem 1, the probability distribution of $U(\theta)$ is equivalent to

$$1 + \frac{1}{2^L} \sum_{i_1,\ldots,i_L=1}^{2} Z_{(1,0,\ldots,0,-1),,\prod_{s=1}^{L} P_s^{i_s} W_s} + \frac{1}{4^L} \sum_{|\lambda|=0,\ |\lambda_+|=2} \sum_{i_1,\ldots,i_L=1}^{4} Z_{\lambda,,\prod_{s=1}^{L} P_s^{i_s} W_s}. \tag{S59}$$

□

IV. Trainability Classification via Zonal Polynomial Decomposition and Its Application to Unitary 1-Designs

A. Proof of Theorem 2

In this subsection, our task is to prove the following theorem:

Theorem 2. *Given a $\mathbb{S}^1$-invariant probability measure μ_R, the i-th-order variance admits the following asymptotic bounds:*

$$\mathrm{Var}_i^{(\mu_L,\mu_R)}(\rho, O) \in O(d^{i-2}) \quad \text{for } i = 0, 1, 2. \tag{S60}$$

Moreover, if ρ is a pure state, then the total variance satisfies the following upper bound:

$$\frac{\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O)}{\int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2]} \leq \frac{d+2}{2} \frac{\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O)}{\int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2]} + \frac{1}{d}. \tag{S61}$$

Proof. As a consequence of classical unitary design theory, it is known that the total variance

$$\mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O) = \frac{1}{d^2-1}\left(\mathrm{Tr}[\rho^2] - \mathrm{Tr}[\rho]^2\right) \int \mathrm{d}\mu_L\,\mathrm{Tr}[X_L^2] \in O(d^{-2}). \tag{S62}$$

In the next step, we will quantify the first- and second-order variances.

We review that a balanced polynomial f of degree t can be expressed as

$$f = \sum_{|\lambda|=0,|\lambda_+|\leq t} \int \mathrm{d}\nu(V) Z_{\lambda,V}, \tag{S63}$$

where ν is a finite signal measure. Here we focus on the first variance and thus take $t = 1$. Consequently, f simplifies to

$$f = 1 + \int \mathrm{d}\nu(V) Z_{(1,0,\cdots,0,-1),V}, \tag{S64}$$

where ν ensures that $f \geq 0$. For brevity, we denote $Z_V := Z_{(1,0,\cdots,0,-1),V}$ throughout this proof. It gives:

$$\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) = \int \mathrm{d}\mu_L(U_L) \int \mathrm{d}\nu(V) \int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger X_L]^2 Z_V(U). \tag{S65}$$

The inner integral can be evaluated using the Weingarten calculus, yielding

$$\begin{aligned}\int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger X_L]^2 Z_V(U) &= \frac{2}{d^2-4}\left(d\,\mathrm{Tr}[\rho^2 V^\dagger X_L^2 V] - 2\,\mathrm{Tr}[\rho]\,\mathrm{Tr}[\rho V^\dagger X_L^2 V] - (\mathrm{Tr}[\rho^2] - \frac{2}{d}\,\mathrm{Tr}[\rho]^2)\,\mathrm{Tr}[X_L^2]\right) \\ &= \frac{2}{d^2-4}\left[d\left(\mathrm{Tr}[\rho^2 V^\dagger X_L^2 V] - \frac{1}{d}\,\mathrm{Tr}[\rho^2]\,\mathrm{Tr}[X_L^2]\right) - 2\,\mathrm{Tr}[\rho]\left(\mathrm{Tr}[\rho V^\dagger X_L^2 V] - \frac{1}{d}\,\mathrm{Tr}[\rho]\,\mathrm{Tr}[X_L^2]\right)\right].\end{aligned} \tag{S66}$$

According to Eq. (S66), the first-order variance can be written as $\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) = S_1 - S_2 + S_3 - S_4$ with

$$\begin{aligned}|S_1| &\leq \int \mathrm{d}\mu_L(U_L) \frac{2d}{d^2-4} |\,\mathrm{Tr}[V\rho^2 V^\dagger X_L^2]| \leq \frac{2d}{d^2-4}\,\mathrm{Tr}[\rho^2] \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2] \in O(d^{-1}), \\ |S_2| &= \frac{2}{(d^2-4)}\,\mathrm{Tr}[\rho^2] \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2] \in O(d^{-2}), \\ |S_3| &\leq \int \mathrm{d}\mu_L(U_L) \frac{4}{d^2-4} |\,\mathrm{Tr}[V\rho V^\dagger X_L^2]| \leq \frac{4}{d^2-4} \int \mathrm{d}\mu_L(U_L) \sqrt{\mathrm{Tr}[\rho^2]\,\mathrm{Tr}[X_L^4]} \\ &\leq \frac{4}{d^2-4}\sqrt{\mathrm{Tr}[\rho^2]} \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2] \in O(d^{-2}), \\ |S_4| &= \frac{4\,\mathrm{Tr}[\rho]^2}{d(d^2-4)} \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2] \in O(d^{-3}).\end{aligned} \tag{S67}$$

Therefore, $\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) \in O(d^{-1})$. Now we introduce the following lemma to proceed in Eq. (S61):

Lemma S.1. *For any complex matrix A and B such that $\mathrm{Tr}(A)\,\mathrm{Tr}(B) = 0$, the function $f(U) = \mathrm{Tr}(AU^\dagger BU)$ is the fixed point of the kernel $K_\nu(f) := \int \mathrm{d}\mu_H(U) Z_V(U) f(U)$, i.e.,*

$$\int \mathrm{d}\mu_H(U) Z_V(U) f(U) = f(V). \tag{S68}$$

By Lemma S.1, for arbitrary complex matrices A and B, we obtain

$$\begin{aligned}\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[AU^\dagger BU] Z_U(V) &= \int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[AU^\dagger BU] Z_V(U) \\ &= \int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[(A - \frac{1}{d}\,\mathrm{Tr}[A] + \frac{1}{d}\,\mathrm{Tr}[A])U^\dagger BU] Z_V(U) \\ &= \mathrm{Tr}[(A - \frac{1}{d}\,\mathrm{Tr}[A])V^\dagger BV] \\ &= \mathrm{Tr}[AV^\dagger BV] - \frac{1}{d}\,\mathrm{Tr}[A]\,\mathrm{Tr}[B].\end{aligned} \tag{S69}$$

Thus, Equation (S66) can be written as

$$\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger X_L]^2 Z_V(U) = \frac{2}{d^2-4} \int \mathrm{d}\mu_H(U) \left(d\,\mathrm{Tr}[U\rho^2 U^\dagger X_L^2] - 2\,\mathrm{Tr}[\rho]\,\mathrm{Tr}[U\rho U^\dagger X_L^2]\right) Z_V(U). \tag{S70}$$

Substituting this into Equation (S65) gives

$$\begin{aligned}\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) &= \frac{2}{d^2-4} \int \mathrm{d}\mu_L(U_L)\,\mathrm{d}\nu(V)\,\mathrm{d}\mu_H(U) \left(d\,\mathrm{Tr}[U\rho^2 U^\dagger X_L^2] - 2\,\mathrm{Tr}[\rho]\,\mathrm{Tr}[U\rho U^\dagger X_L^2]\right) Z_V(U) \\ &= \frac{2}{d^2-4} \int \mathrm{d}\mu_L(U_L) d(\mu_R - \mu_H)(U) \left(d\,\mathrm{Tr}[U\rho^2 U^\dagger X_L^2] - 2\,\mathrm{Tr}[\rho]\,\mathrm{Tr}[U\rho U^\dagger X_L^2]\right).\end{aligned} \tag{S71}$$

When ρ is a pure state, this simplifies to

$$\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) = \frac{2}{d+2} \int \mathrm{d}\mu_L(U_L)\,\mathrm{d}(\mu_R - \mu_H)(U)\,\mathrm{Tr}[U\rho U^\dagger X_L^2], \tag{S72}$$

Since for an Hermitian matrix M and a quantum state $|\psi\rangle$, we have $\langle\psi| M |\psi\rangle^2 \leq \langle\psi| M^2 |\psi\rangle$. It leads to

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) \leq \int \mathrm{d}\mu_L(U_L)\,\mathrm{d}\mu_R(U)\,\mathrm{Tr}[U\rho U^\dagger X_L^2] = \frac{d+2}{2}\mathrm{Var}_1^{(\mu_L,\mu_R)}(\rho, O) + \frac{1}{d}\int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[X_L^2]. \tag{S73}$$

This is just equivalent to Eq. (S61). □

B. Proof of Proposition 1

Proposition 1. *Let ν_m be a probability measure on the unitary group $U(d)$ whose derivative satisfies*

$$\frac{\mathrm{d}\nu_m}{\mathrm{d}\mu_H}(U) = c_m\,\mathrm{Tr}[U\rho U^\dagger P]^{2m}, \tag{S74}$$

where ρ is a pure state, P is an idempotent matrix, and c_m is the normalization constant. Then ν_m defines a unitary 1-design. Moreover, for sufficiently large m, there exists a measure μ_L and an observable O such that

$$\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O) \geq \gamma d^{-1} \tag{S75}$$

for some constant $\gamma > 0$.

Proof. Firstly, we'll prove that the measure ν_m given in Proposition 1 is indeed a unitary 1-design. By definition, a measure μ is a unitary 1-design if and only if for any $f \in \mathrm{Hom}(U(d), 1, 1)$, satisfying

$$\int \mathrm{d}\mu(U) f(U) = \int \mathrm{d}\mu_H(U) f(U). \tag{S76}$$

Notice that $Hom(U(d), 1, 1)$ is spanned by the basis 1 and $Z_{(1,0,\cdots,0,-1),V} = (d^2-1)(|\,\mathrm{V}^\dagger\cdot|^2 - 1)$. Thus, the problem reduces to proving that the integral of the product of $Z_{(1,0,\cdots,0,-1)}$ and the variance function is 0, i.e.

$$\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger P]^{2m}|\,\mathrm{Tr}[U^\dagger V]|^2 = \int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger P]^{2m}. \tag{S77}$$

Expanding the left side of (S77) by the Einstein summation convention, it gives

$$\begin{aligned}&\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger P]^{2m}|\,\mathrm{Tr}[U^\dagger V]|^2 = \\ &\sum_{\substack{i_1,i_2\cdots,i_{2n},i_{2n+1}\\ i'_1,i'_2,\cdots,i'_{2n},i'_{2n+1}\\ j,j'}} \int \mathrm{d}\mu_H(U) U_{i_1,1}U_{i_2,1}\cdots U_{i_{2n},1}U_{i_{2n+1},j}U^*_{i'_1,1}U^*_{i'_2,1}\cdots U^*_{i'_{2n},1}U^*_{i'_{2n+1},j'}\cdot V^*_{i_{2n+1},j}V_{i'_{2n+1},j'}P_{i_1,i'_1}P_{i_2,i'_2}\cdots P_{i_{2n},i'_{2n}}.\end{aligned} \tag{S78}$$

In general, it is hard to calculate such a symbolic integral by Theorem S.1. But here we can apply the idea of tensor network in [1] to capture this problem. In the tensor network associated with the symbolic integral in Eq.(S78), there exists a chain $PP\cdots P$ connecting with $V^\dagger V$, which is shown in Figure S1. Since $V^\dagger V = I$, the connection in Figure S1 is the same as the connection of

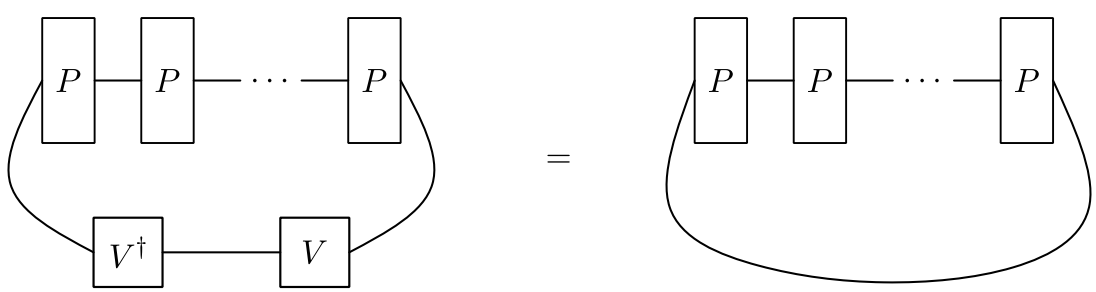

FIG. S1: The Schematic diagram of the integral calculation represented by tensor networks.

$PP\cdots P$ end to end. Thus,

$$\begin{aligned}&\int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U\rho U^\dagger P]^{2m}|\,\mathrm{Tr}[U^\dagger V]|^2 = \\ &\sum_{\substack{i_1,i_2\cdots,i_{2n},i_{2n+1}\\ i'_1,i'_2,\cdots,i'_{2n},i'_{2n+1}}} \int \mathrm{d}\mu_H(U) U_{i_1,1}U_{i_2,1}\cdots U_{i_{2n},1}U_{i_{2n+1},j}U^*_{i'_1,1}U^*_{i'_2,1}\cdots U^*_{i'_{2n},1}U^*_{i'_{2n+1},j'}\cdot P_{i_1,i'_1}P_{i_2,i'_2}\cdots P_{i_{2n},i'_{2n}} \\ &= \int \mathrm{d}\mu_H(U)\,\mathrm{Tr}[U^\rho U^\dagger P]^{2m}.\end{aligned} \tag{S79}$$

Here, we have proved that the measure μ is a unitary 1-design. Thus, the corresponding circuit is a zero-bias circuit. The total variance can be expressed as

$$\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O) = c_m \int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger P]^{2m}\, \mathrm{Tr}[U\rho U^\dagger X]^2. \tag{S80}$$

Using the same method, the result in (S80) is the linear combination of $\mathrm{Tr}[PXPX]$, $\mathrm{Tr}[PX]^2$ and $\mathrm{Tr}[X^2]$. By using basic permutation and combination formulas and the Weingarten summation formula, the coefficient in front of each term can be precisely calculated. Here, to highlight the main calculations of permutations and combinations, all coefficients have been factored out of the factor

$$\sum_{\sigma\in S_{2m}} \mathrm{Wg}(\sigma, d) = \frac{(d-1)!}{(2m+d-1)!}. \tag{S81}$$

Now it is the time to calculate the coefficients of each term:

1. the coefficient of the term $\mathrm{Tr}[PX]^2$:

$$\begin{aligned}\sum_{\substack{k=0\\ k \text{ odd}}}^{2m}\sum_{\substack{l=0\\ l \text{ odd}}}^{2m-k}\binom{2m}{k}\binom{2m-k}{l}k!l!\sum_{\sigma\in S^e_{2m-k-l}} d^{c(\sigma)} &= \sum_{\substack{k=0\\ k \text{ odd}}}^{2m}\sum_{\substack{l=0\\ l \text{ odd}}}^{2m-k}\frac{(2m)!}{(2m-k-l)!}\sum_{\sigma\in S^e_{2m-k-l}} d^{c(\sigma)}\\ &= (2m)!\sum_{p=0}^{m}\frac{m-p}{(2p)!}\sum_{\sigma\in S^e_{2p}} d^{c(\sigma)}.\end{aligned} \tag{S82}$$

   Here, S^e_r represents the subset consisting of all the permutations generated by even circles in S_r.

2. the coefficient of the term $\mathrm{Tr}[PXPX]$:

$$\begin{aligned}\sum_{\substack{s=0\\ s \text{ even}}}^{2m}\binom{2m}{s}\frac{s}{2}s!\sum_{\sigma\in S^e_{2m-s}} d^{c(\sigma)} &= \frac{(2m)!}{2}\sum_{s=0}^{m}\frac{2s}{(2m-2s)!}\sum_{\sigma\in S^e_{2m-2s}} d^{c(\sigma)}\\ &= (2m)!\sum_{s=0}^{m}\frac{m-s}{(2m)!}\sum_{\sigma\in S^e_{2s}} d^{c(\sigma)}.\end{aligned} \tag{S83}$$

3. the coefficient of the term $\mathrm{Tr}[X^2]$:

$$\begin{aligned}\sum_{\substack{s=0\\ s \text{ even}}}^{2m}\binom{2m}{s}\left(\frac{s}{2}+1\right)s!\sum_{\sigma\in S^e_{2m-s}} d^{c(\sigma)} &= \frac{(2m)!}{2}\sum_{s=0}^{m}\frac{2(s+1)}{(2m-2s)!}\sum_{\sigma\in S^e_{2m-2s}} d^{c(\sigma)}\\ &= (2m)!\sum_{s=0}^{m}\frac{m-s+1}{(2m)!}\sum_{\sigma\in S^e_{2s}} d^{c(\sigma)}.\end{aligned} \tag{S84}$$

For the normalized factor c_m, it is similar to obtain

$$1/c_m = \int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger P]^{2m} = \frac{(d-1)!}{(2m+d-1)!}\sum_{\sigma_\in S^e_{2m}} d^{c(\sigma)}. \tag{S85}$$

By Eq.(S82), (S83), (S84)and(S85), the total variance

$$\begin{aligned}\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O) = &\frac{(2m)!}{(2m+d)(2m+d+1)g(m)}\left(\sum_{s=0}^{m}\frac{m-s}{(2s)!}g(s)\right)\Big(\mathrm{Tr}[PX]^2+\mathrm{Tr}[PXPX]\Big)\\ &+\frac{(2m)!}{(2m+d)(2m+d+1)g(m)}\left(\sum_{s=0}^{m}\frac{m-s+1}{(2s)!}g(s)\right)\mathrm{Tr}[X^2].\end{aligned} \tag{S86}$$

For convenience, note that

$$g(s) := \sum_{\sigma \in S_{2s}^{e}} d^{c(\sigma)}. \tag{S87}$$

By the definition, $g(s) \in \Theta(d^s)$. Indeed, there is an explicit form:

$$g(s) = \begin{cases} \dfrac{(2s)!}{2^s s!} \displaystyle\prod_{\ell=0}^{s-1} (d+2\ell), & s \geq 1 \\ 1, & s = 0 \end{cases}. \tag{S88}$$

When $s \geq 1$, Eq.(S88) can be represented by Gamma functions(Here s is an integer. So the Gamma functions are indeed factorials.):

$$\prod_{\ell=0}^{s-1} (d+2\ell) = 2^s \prod_{\ell=0}^{s-1} (d/2+\ell) = 2^s \frac{(d/2+s-1)!}{(d/2-1)!}. \tag{S89}$$

When $s = 0$, Eq.(S89) gives $g(0) = 1$, satisfying Eq.(S88). Thus, for any $s \in \mathbb{Z}_+$, there is a unified expression:

$$g(s) = \frac{(2s)!}{s!} \frac{(d/2+s-1)!}{(d/2-1)!}. \tag{S90}$$

For the first summation,

$$\begin{aligned} \sum_{s=0}^{m} \frac{m-s}{(2s)!} g(s) &= \sum_{s=0}^{m} (m-s) \binom{d/2-1+s}{d/2-1} \\ &= m \sum_{s=0}^{m} \binom{d/2-1+s}{d/2-1} - \sum_{s=0}^{m} s \binom{d/2-1+s}{d/2-1} \\ &= m \sum_{s=0}^{m} \binom{d/2-1+s}{d/2-1} - \frac{d}{2} \sum_{s=0}^{m-1} \binom{d/2+s}{d/2} \\ &= m \binom{d/2+m}{d/2} - \frac{d}{2} \binom{d/2+m}{d/2+1} = \frac{2m}{d+2} \binom{d/2+m}{d/2}. \end{aligned} \tag{S91}$$

where the last equality uses the Hockey-stick identity. For the first summation, applying the same method, we obtain

$$\sum_{s=0}^{m} \frac{m-s+1}{(2s)!} g(s) = \left(1 + \frac{2m}{d+2}\right) \binom{d/2+m}{d/2}. \tag{S92}$$

Substituting Eq.(S90), (S91) and (S92) into Eq.(S86), the final result

$$\mathrm{Var}^{(\mu_L,\nu_m)}(\rho, O) = \frac{2m}{d(2m+d+1)(d+2)} \left(\mathrm{Tr}[PX]^2 + \mathrm{Tr}[PXPX]\right) + \frac{2m+d+2}{d(2m+d+1)(d+2)} \mathrm{Tr}[X^2]. \tag{S93}$$

A suitable choice of P and X can make $\mathrm{Tr}[PX]^2$ or $\mathrm{Tr}[PXPX]$ achieve the level of $\Theta(d)\,\mathrm{Tr}[X^2]$. Thus, when m is sufficiently large, the total variance in Eq. (S93) can achieve the level of $\Theta(d^{-1})\,\mathrm{Tr}[X^2]$ through choosing suitable P and X. □

V. Improvement Potential of Barren Plateaus

A. Proof of Theorem 3

In this section, we will prove Theorem 3 in the main text, which is stated as follows:

Theorem 3. *For a PQC at θ_ℓ with the probability measure pair (μ_L, μ_R), the improvement potential obtained by inserting an oracle is*

$$k_{\mu_R} = \frac{\|g * h\|_\infty}{\|gh\|_1}. \tag{S94}$$

Here g and h are functions on $U(d)$ defined by

$$g(V) = \frac{\mathrm{d}\mu_R}{\mathrm{d}\mu_H}(V), h(V) = \int \mathrm{d}\mu_L(U_L)\ \mathrm{Tr}\left[V^\dagger \rho V X_L\right]^2. \tag{S95}$$

The function g denotes the derivative of μ_R with respect to μ_H, while h represents the variance obtained after integrating over the left unitary. Both functions can be decided by the pair (μ_L, μ_R).

Proof. For $\mathrm{d}\mu(U_R) = g(U_R)\,\mathrm{d}\mu_H(U_R)$, the variance is of the form

$$\begin{aligned}\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) &= \int \mathrm{d}\mu_L(U_L) \int \mathrm{d}\mu_H(U_R) g(U_R)\,\mathrm{Tr}[U_R \rho U_R^\dagger X_L]^2 \\ &= \int \mathrm{d}\mu_H(V) g(V) h(V^\dagger),\end{aligned} \tag{S96}$$

where $h(U) = \int d\mu_L(U_L)\,\mathrm{Tr}[U^\dagger \rho U X_L]^2$ is a homogeneous balanced polynomial of degree 2. Since ρ and X_L are Hermitian, $\mathrm{Tr}[U^\dagger \rho U X_L] \in \mathbb{R}$ for all $U \in U(d)$. Thus,

$$\mathrm{Tr}[U\rho U^\dagger X_L] = \overline{\mathrm{Tr}[U\rho U^\dagger X_L]} = \mathrm{Tr}[U^\dagger \rho U X_L]. \tag{S97}$$

Then we have $h(U) = h(U^\dagger)$, so we can simplify the form of the variance

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = g * h(I) = \|gh\|_1. \tag{S98}$$

Now we insert an oracle of probability measure μ with the derivative $\frac{\mathrm{d}\mu}{\mathrm{d}\mu_H} = f$. The variance has the form

$$\begin{aligned}\mathrm{Var}^{(\mu_L,\mu*\mu_R)}(\rho, O) &= \int \mathrm{d}\mu_L(U_L) \int \mathrm{d}\mu_H(U_R)(f * g)(U_R)\,\mathrm{Tr}[U_R \rho U_R^\dagger X_L]^2 \\ &= \int \mathrm{d}\mu_H(U_R)\,\mathrm{d}\mu_H(V) f(V) g(V^\dagger U_R) \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[U_R \rho U_R^\dagger X_L]^2 \\ &= \int \mathrm{d}\mu_H(V) f(V^\dagger) \int \mathrm{d}\mu_H(U_R) g(VU_R) \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[U_R \rho U_R^\dagger X_L]^2 \\ &= \int \mathrm{d}\mu_H(V) f(V^\dagger) \int \mathrm{d}\mu_H(U_R) g(U_R) \int \mathrm{d}\mu_L(U_L)\,\mathrm{Tr}[V^\dagger U_R \rho U_R^\dagger V X_L]^2 \\ &= \int \mathrm{d}\mu_H(V) f(V^\dagger)(g * h)(V).\end{aligned} \tag{S99}$$

Thus, the improvement potential

$$k_{\mu_R} = \sup_{\mu \in \mathcal{P}(\mathcal{U}(d))} \frac{\mathrm{Var}^{(\mu_L,\mu*\mu_R)}(\rho, O)}{\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O)} = \sup_f \frac{\|f^\dagger(g * h)\|_1}{\|gh\|_1}, \tag{S100}$$

where $f^\dagger(U) = f(U^\dagger)$. By Hölder's inequality,

$$\|f^\dagger(g * h)\|_1 \le \|f^\dagger\|_1 \|g * h\|_\infty, \tag{S101}$$

and the equality holds if and only if $|g * h| = \|g * h\|_\infty$ almost everywhere on $\mathrm{supp}(f)$. Let $\mu_H(\mathrm{supp}(f)) \to 0$, preserving $\|f\|_1 = 1$ at the same time. The process makes $\|f^\dagger(g * h)\|_1 \to \|g * h\|_\infty$. Thus, we obtain

$$k_{\mu_R} = \frac{\|g * h\|_\infty}{\|gh\|_1}. \tag{S102}$$

Indeed, $\|g * h\|_\infty = \max_{V \in \mathcal{U}(d)} \|g_V h\|_1$. □

In the following subsections V B, V C, we will provide several examples that clearly determine whether it has the improvement potential. In the subsection V D, we will show the attainable maximum value of the improvement potential.

B. PQCs with Small Variances must Have an Improvement Potential

Proposition S.4. *If the right measure μ_R leads to a smaller variance than a unitary 2-design, i.e.,*

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) < \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O), \tag{S103}$$

then the improvement potential $k_{\mu_R} > 1$. Moreover, if the right measure forms a unitary 2-design, then $k_{\mu_R} = 1$.

Proof. If the variance

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) < \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O), \tag{S104}$$

we assume that $k_{\mu_R} = 1$, i.e. $\|g * h\|_\infty = (g * h)(I)$, then

$$\|g * h\|_\infty < \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O) = \|h\|_1 = \|g * h\|_1. \tag{S105}$$

However, this violates Hölder's inequality. Thus, $k_{\mu_R} > 1$.

Now we assume that μ forms a unitary 2-design, then

$$\|g * h\|_\infty = \|h\|_1 = \|gh\|_1 \tag{S106}$$

by definition. According to Theorem 3, $k_{\mu_R} = 1$. □

C. Examples of PQCs without an Improvement Potential

Proposition S.5. *We assume that U_L forms a unitary 2-design μ_L and U_R follows a probability measure μ_R whose derivative (with respect to the Haar measure) is a balanced polynomial of degree 1. Then,*

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O), \tag{S107}$$

which implies the improvement potential $k_{\mu_R} = 1$.

Proof. By Eq.(S5) and Eq.(S6), one can calculate that

$$\begin{aligned}
\mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O) &= \int \mathrm{d}\mu_H(U_L)\, \mathrm{d}\mu_H(U_R)\, \mathrm{Tr}[U\rho U^\dagger X_L]^2 \\
&= \int \mathrm{d}\mu_H(U_L) \frac{1}{d^2-1}\left(\mathrm{Tr}[\rho^2] - \frac{1}{d}\mathrm{Tr}[\rho]^2\right)\mathrm{Tr}[X_L^2] \\
&= \frac{2d^2}{(d^2-1)^2}\left(\mathrm{Tr}[\rho^2] - \frac{1}{d}\mathrm{Tr}[\rho]^2\right)\left(\mathrm{Tr}[O^2] - \frac{1}{d}\mathrm{Tr}[O]^2\right)
\end{aligned} \tag{S108}$$

and

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = \frac{1}{d^2-1}\left(\mathrm{Tr}[O^2] - \frac{1}{d}\mathrm{Tr}[O]^2\right)\int \mathrm{d}\mu_R(U_R)\,\mathrm{Tr}[Y_L^2]. \tag{S109}$$

Here, $Y_L = i[U_R\rho U_R^\dagger, H_\ell]$, which leads that

$$\int \mathrm{d}\mu_R(U_R)\,\mathrm{Tr}[Y_L^2] = \int \mathrm{d}\mu_R(U_R) 2(\mathrm{Tr}[\rho^2] - \mathrm{Tr}[(U_R\rho U_R^\dagger H_\ell)^2]) = 2\,\mathrm{Tr}[\rho^2] - 2\int \mathrm{d}\mu_R(U_R)\,\mathrm{Tr}[(U_R\rho U_R^\dagger H_\ell)^2]. \tag{S110}$$

Since $\mathrm{d}\mu_R/\mathrm{d}\mu_H$ is a balanced polynomial of degree 1, we can write it in the form

$$\frac{\mathrm{d}\mu_R}{\mathrm{d}\mu_H}(U) = \sum_{i,j,i',j'} R^{i,j}_{i',j'} U_{i,j} U^*_{i',j'} + C, \tag{S111}$$

where $R^{i,j}_{i',j'}$ and C are real such that μ_R is a probability measure. Using the form in Eq. (S111), the latter term in Eq. (S110) can be calculated by

$$\begin{aligned}
\int \mathrm{d}\mu_R(U_R)\,\mathrm{Tr}[(U_R\rho U_R^\dagger H_\ell)^2] &= \int \mathrm{d}\mu_H(U)\left(\sum_{i,j,i',j'} R^{i,j}_{i',j'} U_{i,j} U^*_{i',j'} + C\right)\mathrm{Tr}[(U\rho U^\dagger H_\ell)^2] \\
&= \int \mathrm{d}\mu_H(U)\left(\sum_{i,j,i',j'} R^{i,j}_{i',j'} U_{i,j} U^*_{i',j'}\right)\mathrm{Tr}[(U\rho U^\dagger H_\ell)^2] + \frac{Cd}{d^2-1}(\mathrm{Tr}[\rho]^2 - \frac{1}{d}\mathrm{Tr}[\rho^2])\,.
\end{aligned} \tag{S112}$$

The formal term can be calculated by a symbolic integral,

$$
\begin{aligned}
&\int \mathrm{d}\mu_H(U)\Big(\sum_{i,j,i',j'} R^{i,j}_{i',j'} U_{i,j} U^*_{i',j'}\Big)\,\mathrm{Tr}[(U\rho U^\dagger H_\ell)^2] \\
&\qquad = \int \mathrm{d}\mu_H(U) \sum U_{i_1,j_1} U_{i_2,j_2} U_{i_3,j_3} U^*_{i'_1,j'_1} U^*_{i'_2,j'_2} U^*_{i'_3,j'_3} R^{i_1,j_1}_{i'_1,j'_1} \rho_{j_2,j'_2} \rho_{j_3,j'_3} (H_\ell)_{i'_2,i_2} (H_\ell)_{i'_3,i_3} \\
&\qquad = (d\mathrm{Wg}(1^3,d) + 2\mathrm{Wg}(21,d)) \left(\sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1}\right) \mathrm{Tr}[H_\ell^2] \\
&\qquad + (d\mathrm{Wg}(21,d) + \mathrm{Wg}(1^3,d) + \mathrm{Wg}(3,d)) \left(\sum_{i_1,i'_1} R^{i_1,i'_1}_{i'_1,i_1} \mathrm{Tr}[H_\ell^2] + \sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1}\right) \\
&\qquad + (d\mathrm{Wg}(21,d) + 2\mathrm{Wg}(3,d)) \left(\sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1}\right) \\
&\qquad + (d\mathrm{Wg}(3,d) + 2\mathrm{Wg}(21,d)) \left(\sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1} + \sum_{i_1,i'_1} R^{i_1,i'_1}_{i_1,i'_1}\right) \\
&\qquad = \frac{1}{d^2-1} \left(\sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1}\right) \left(\mathrm{Tr}[\rho]^2 - \frac{1}{d}\mathrm{Tr}[\rho^2]\right).
\end{aligned}
\tag{S113}
$$

Since μ_R is a probability measure, the restriction on the integral leads to

$$
1 = \int \mathrm{d}\mu_H(U) \sum_{i_1,j_1,i'_1,j'_1} R^{i_1,j_1}_{i'_1,j'_1} U_{i_1,j_1} U^*_{i'_1,j'_1} + C = \frac{1}{d} \sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1} + C. \tag{S114}
$$

By Eq.(S112),(S113) and (S114), we obtain

$$
\begin{aligned}
\int \mathrm{d}\mu_R(U_R)\,\mathrm{Tr}[Y_L^2] &= 2\left(\mathrm{Tr}[\rho^2] - \frac{1}{d^2-1}\left(\sum_{i_1,i'_1} R^{i_1,i_1}_{i'_1,i'_1} + Cd\right)\left(\mathrm{Tr}[\rho^2] - \frac{1}{d}\mathrm{Tr}[\rho]^2\right)\right) \\
&= \frac{2d^2}{d^2-1}\left(\mathrm{Tr}[\rho^2] - \frac{1}{d}\mathrm{Tr}[\rho]^2\right).
\end{aligned}
\tag{S115}
$$

Therefore, the variance of (μ_L, μ_R) has the explicit form:

$$
\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = \frac{2d^2}{(d^2-1)^2}\left(\mathrm{Tr}[\rho^2] - \frac{1}{d}\mathrm{Tr}[\rho]^2\right)\left(\mathrm{Tr}[O^2] - \frac{1}{d}\mathrm{Tr}[O]^2\right). \tag{S116}
$$

Since μ_R is a balanced polynomial measure of degree 1, any convolution measure $\mu * \mu_R$ is a balanced polynomial measure of degree up to 1. Thus,

$$
\mathrm{Var}^{(\mu_L,\mu*\mu_R)}(\rho, O) = \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O) = \mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O), \tag{S117}
$$

meaning that $k_{\mu_R} = 1$. □

D. Examples of PQCs with the Improvement Potential Beyond Constant Scaling

Proposition 2. *Assume that the left unitary matrix U_L commutes with the observable O, and μ_R is a polynomial probability measure with the following derivative:*

$$
\frac{\mathrm{d}\mu_R}{\mathrm{d}\mu_H}(U) = \binom{m+d-1}{d-1} \mathrm{Tr}[U\rho U^\dagger |\psi\rangle\langle\psi|]^{2m}, \tag{S118}
$$

where ρ is a pure state. Further assume that $H_\ell = P_1$ and $O = P_2/\sqrt{d}$, where P_1 and P_2 do not commute with each other. Then there exists a quantum state $|\phi\rangle$ such that

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) > \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O), \tag{S119}$$

and when n is sufficiently large, the improvement potential of (μ_L, μ_R) can reach the order of d.

Proof. According to the given conditions, and following the notation of Theorem 3, we have

$$\|g * h\|_1 = \binom{m+d-1}{d-1} \int \mathrm{d}\mu_L(U_L)\mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger X_L]^2\, \mathrm{Tr}[U\rho U^\dagger |\psi\rangle\langle\psi|]^{2m}. \tag{S120}$$

Therefore, the problem reduces to computing the following integral

$$\int \mathrm{d}\mu_H(U)\, \mathrm{Tr}[U\rho U^\dagger X_L]^2\, \mathrm{Tr}[U\rho U^\dagger |\psi\rangle\langle\psi|]^{2m}. \tag{S121}$$

Since ρ is a pure state and $|\psi\rangle$ is an eigenvector of X_L, Eq. (S121) can be simplified to

$$\int \mathrm{d}\mu_H(U) \sum_{i_1,i_2,i_1',i_2'} U_{i_1 1}U_{i_1' 1}U_{i_2 1}U_{i_2' 1}|U_{11}|^{2n} D(X_L)_{i_1' i_1} D(X_L)_{i_2' i_2}. \tag{S122}$$

Here, $D(X_L)$ is the diagonalized matrix of X_L. For convenience, we let the first index of $D(X_L)$ correspond to $|\psi\rangle$. According to Theorem S.1 and Eq. (S7), using the symbolic integration method, we perform the following calculation:

$$\begin{aligned}
&\int \mathrm{d}\mu_H(U) \sum_{i_1,i_2,i_1',i_2'} U_{i_1 1}U_{i_1' 1}U_{i_2 1}U_{i_2' 1}|U_{11}|^{2n} D(X_L)_{i_1' i_1} D(X_L)_{i_2' i_2} \\
&= \sum_{i_1,i_2,i_1',i_2'} \sum_{\sigma,\tau\in S_{m+2}} \mathrm{Wg}(\sigma\tau^{-1}, d)\delta_{i_1,i'_{\sigma(1)}}\delta_{i_2,i'_{\sigma(2)}}\cdots\delta_{i_{m+2},i'_{\sigma(m+2)}} D(X_L)_{i_1' i_1} D(X_L)_{i_2' i_2} \\
&= \sum_{i_1,i_2,i_1',i_2'} \sum_{\sigma,\tau\in S_{m+2}} \mathrm{Wg}(\tau, d)\delta_{i_1,i'_{\sigma(1)}}\delta_{i_2,i'_{\sigma(2)}}\cdots\delta_{i_{m+2},i'_{\sigma(m+2)}} D(X_L)_{i_1' i_1} D(X_L)_{i_2' i_2} \\
&= \frac{(d-1)!}{(m+d+1)!} \sum_{i_1,i_2,i_1',i_2'} \delta_{i_1,i'_{\sigma(1)}}\delta_{i_2,i'_{\sigma(2)}}\cdots\delta_{i_{m+2},i'_{\sigma(m+2)}} D(X_L)_{i_1' i_1} D(X_L)_{i_2' i_2} \\
&= \frac{(d-1)!}{(m+d+1)!}(|S_m|\,\mathrm{Tr}[X_L^2] + m(m-1)|S_m|\langle\psi|X_L|\psi\rangle^2 + 2m|S_m|\langle\psi|X_L^2|\psi\rangle) \\
&= \frac{m!(d-1)!}{(m+d+1)!}(\mathrm{Tr}[X_L^2] + m(m-1)\langle\psi|X_L|\psi\rangle^2 + 2m\langle\psi|X_L^2|\psi\rangle).
\end{aligned} \tag{S123}$$

Here, the specific process of symbolic integration is omitted, and the third equality uses the summation formula for the Weingarten function. Finally, we obtain

$$\|gh\|_1 = \frac{1}{(m+d)(m+d+1)} \int \mathrm{d}\mu_L(U_L) \left(\mathrm{Tr}[X_L^2] + m(m-1)\langle\psi|X_L|\psi\rangle^2 + 2m\langle\psi|X_L^2|\psi\rangle\right). \tag{S124}$$

It should be noted that replacing $|\psi\rangle$ given in the problem statement with an arbitrary eigenvector of X_L does not affect the result given by Eq. (S124). Therefore, according to Theorem 3, the improvement potential of this network is

$$k_{\mu_R} = \frac{\|g * h\|_\infty}{\|gh\|_1} = \frac{\max_{V\in U(d)} \|g_V h\|_1}{\|gh\|_1} \geq \frac{\max_{V\in U(d)} \int \mathrm{d}\mu_L(U_L) \left(\mathrm{Tr}[X_L^2] + m(m-1)\langle\psi|X_{L,V}|\psi\rangle^2 + 2m\langle\psi|X_{L,V}^2|\psi\rangle\right)}{\int \mathrm{d}\mu_L(U_L) \left(\mathrm{Tr}[X_L^2] + m(m-1)\langle\psi|X_L|\psi\rangle^2 + 2m\langle\psi|X_L^2|\psi\rangle\right)}. \tag{S125}$$

where $X_{L,V} = VX_LV^\dagger$. Substituting $H_\ell = P_1$ and $O = P_2/\sqrt{d}$, and noting $[P_1, P_2] = 2iP$, we have $X_L = -2P$. Consequently,

$$k_\mu \geq \frac{4 + 8m/d + 4m(m-1)/d \max_{V\in U(d)} \left\langle\psi|VPV^\dagger|\psi\right\rangle^2}{4 + 8m/d + 4m(m-1)/d\,\langle\psi|P|\psi\rangle^2} = \frac{4 + 8m/d + 4m(m-1)/d}{4 + 8m/d + 4m(m-1)/d\,\langle\psi|P|\psi\rangle^2}. \tag{S126}$$

Here, we choose $|\psi\rangle$ such that $2/(d+1) < \langle\psi|P|\psi\rangle^2 < 3/(d+1)$. Since P is a Pauli string and $\{\langle\psi|P|\psi\rangle^2 : \psi \in \mathbb{CP}^n\} = [0,1]$, such a choice of $|\psi\rangle$ is valid. This leads, for sufficiently large m, to

$$\frac{d+1}{3} < k_\mu < \frac{d+1}{2}. \tag{S127}$$

Meanwhile,

$$\mathrm{Var}^{(\mu_L,\mu_R)}(\rho, O) = \|gh\|_1 > \mathrm{Var}^{(\mu_L,\mu_H)}(\rho, O) \tag{S128}$$

satisfies the conditions given in the proposition. □

VI. Numerical Simulations on Polynomial Measures

In this section, we will state the derivations and proofs for numerical simulations. We will begin with the foundation for our circuit setting. All Pauli matrices in the rotation gates in our circuit are taken from the Gell-Mann Basis [14].

A. Gell-Mann Basis for the Lie Algebra $\mathfrak{su}(N)$

The Gell-Mann type basis for $\mathfrak{su}(N)$ consists of N^2-1 traceless Hermitian matrices, which are defined as follows:

1. For every $i, j = 1, 2, \cdots, N; i < j$, we define two $N \times N$ matrices

$$\begin{aligned} [\lambda^1_{i,j}]_{\mu\nu} &= \delta_{j\mu}\delta_{i\nu} + \delta_{j\nu}\delta_{i\mu} \\ [\lambda^2_{i,j}]_{\mu\nu} &= -i\delta_{j\mu}\delta_{i\nu} + \delta_{j\nu}\delta_{i\mu} \end{aligned}, \tag{S129}$$

   which form $N(N-1)$ linearly independent matrices.

2. Another $N-1$ diagonal matrices are as follows:

$$\mu_n = \sqrt{\frac{2}{n(n+1)}} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & -n & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \tag{S130}$$

   for $n = 1, 3, \cdots, N-1$.

B. Construction of Balanced Polynomial Measures

For the case of binary qubits, the Euler angle parametrization for $U(4)$ was initially given in [13] by

$$U = e^{i\alpha_0} e^{i\lambda_3\alpha_1} e^{i\lambda_2\alpha_2} e^{i\lambda_3\alpha_3} e^{i\lambda_5\alpha_4} e^{i\lambda_3\alpha_5} e^{i\lambda_{10}\alpha_6} e^{i\lambda_3\alpha_7} e^{i\lambda_2\alpha_8} e^{i\lambda_3\alpha_9} e^{i\lambda_5\alpha_{10}} e^{i\lambda_3\alpha_{11}} e^{i\lambda_2\alpha_{12}} e^{i\lambda_3\alpha_{13}} e^{i\lambda_8\alpha_{14}} e^{i\lambda_{15}\alpha_{15}}, \tag{S131}$$

Here, $\{\lambda_i\}_{i=1}^{15}$ denotes the set consisting of the Gell-Mann basis for the Lie algebra $\mathfrak{su}(4)$[6]. The Haar measure on $U(4)$, described in [13, 14], is parameterized by angles as follows:

$$\mathrm{d}\mu_H(U) = \frac{144\sqrt{2}}{\pi^{10}} \sin(2\alpha_2)\cos^3(\alpha_4)\sin(\alpha_4)\cos(\alpha_6)\sin^5(\alpha_6)\sin(2\alpha_8)\cos(\alpha_{10})\sin^3(\alpha_{10})\sin(2\alpha_{12})\,\mathrm{d}\alpha_{15}\cdots\mathrm{d}\alpha_1\,\mathrm{d}\alpha_0. \tag{S132}$$

For example, consider a homogeneous balanced polynomial probability measure μ with the derivative

$$\frac{\mathrm{d}\mu}{\mathrm{d}\mu_H}(U) = 4|U_{44}|^2, \tag{S133}$$

where $U_{44} = e^{-i\sqrt{\frac{3}{2}}\lambda_{15}}\cos(\alpha_6)$ is parametrized by the Euler angles. One can also check that μ is a probability measure through the symbolic integral. Let $\alpha_0, \alpha_1, \cdots, \alpha_{15}$ follow the distribution in Table S2, then the corresponding PQC is associated with the probability measure μ.

Parameters	Distributions	Domains
α_0	Uniform	$[0, 2\pi)$
$\alpha_1, \alpha_3, \alpha_5, \alpha_7, \alpha_9, \alpha_{11}, \alpha_{13}$	Uniform	$[0, \pi]$
α_{14}	Uniform	$[0, \pi/\sqrt{3}]$
α_{15}	Uniform	$[0, \pi/\sqrt{6}]$
$\alpha_2, \alpha_8, \alpha_{12}$	$P(x) = \sin(2x)$	$[0, \pi/2]$
α_4	$P(x) = 4\cos^3(x)\sin(x)$	$[0, \pi/2]$
α_6	$P(x) = 24\cos^3(x)\sin^5(x)$	$[0, \pi/2]$
α_{10}	$P(x) = 4\cos(x)\sin^3(x)$	$[0, \pi/2]$

TABLE S2: Parameters for the PQC satisfying probability measure μ in the binary qubits system.

C. Circuit Ansatz Setting

We have given a quantum circuit setup in the main text, as shown in Figure 4 of the main text. Here, we give all the settings in detail. In our circuit setting of right unitaries, G_k and D_k represent rotation gates of different structures. Here, $G_k = \prod_{j=1}^{k} \exp\big(iT\theta_{k,2j-1}\big)\exp\big(iS_j\theta_{k,2j}\big)$ and $D_k = \exp(iM_k\alpha_k)$. According to the Gell-Mann type basis for $\mathfrak{su}(d)$ and the Euler angle parametrization for $U(d)$ given in [14], we choose

$$T = \mu_1, \quad S_j = \lambda^1_{1,j}, \quad M_j = \mu_j. \tag{S134}$$

For the left part, we take $U_L = \exp(iP\beta)$ for some Pauli word P and $\beta \sim \text{Uniform}[0, 2\pi)$. Making the left part simple, as a single rotation gate, can highlight the role that the right unitary structure plays. By the way, since β satisfying the uniform distribution, we have

$$\mathbb{E}_\theta^{(\mu_L,\mu_R)}[\partial_{\alpha_{d-1}} C] = 0. \tag{S135}$$

Thus, we can apply

$$\text{Var}_\theta^{(\mu_L,\mu_R)}[\partial_\ell C] = \mathbb{E}_\theta^{(\mu_L,\mu_R)}[(\partial_{\alpha_{d-1}} C)^2] \tag{S136}$$

in the calculation of the simulation results.

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